UNIVERSITY OF CENTRAL PUNJAB

(Faculty of IT)

Computer Communications and Networks (CCN)

Assignment No 2 **Date : 04 Apr 2026**

Total Marks: 100 **Due Date : 13 Apr 2026**

CLO 2 Questions

1. Draw the flow chart for concurrent TCP server communicating with a TCP client. Why do we use send() and recv() calls in TCP client-server, and not the sendto() and recvfrom() calls.

2. Consider two hosts A & B connected by a single link of rate **R** bps. Suppose the two hosts are separated by **m** meters and propagation speed along the link is **s** meters/sec. Host A is to send a packet of length **L** bits to host B. Answer the following:

   (a) Suppose host A begins to transmit packet at time t=0. At time t= d_{trans} , where is the last bit of the packet.
   (b) Suppose d_{prop} is greater than d_{trans} . At time t= d_{trans} , where is the first bit of the packet.
   (c) Suppose d_{trans} is greater than d_{prop}. At time t= d_{trans} , where is the first bit of the packet.
   (d) Suppose s = 2.5×10^8 m/s, L=100 bits, and R=28 kbps. Find the distance m such that $d_{prop} = d_{trans}$

3. What is the difference between HTTP GET and HTTP Conditional GET; why the later is used?

4. An HTTP client requests a web page from an HTTP server. The web page (base html page) references 7 more objects; 03 located on the same server, and 02 each at two other servers. If average RTT is 4 msec, i.e. RTT for connection establishment, or for getting one object, and average transmission time for each object is 10 msec, compute the delay involved to download the complete web page and all the objects one by one sequentially, when using persistent HTTP.

5. Suppose you open a startup company "theBest" and want to set up your company network. Your network has the following servers:-

   - Primary and Secondary Authoritative DNS Servers dns1.theBest.com and dns2.theBest.com having IP Addresses "111.19.128.35" and "111.19.128.36" respectively.
   - Web Server www.theBest.com having IP Address of 111.19.128.45
   - Email Server mail.theBest.com having IP Address of 111.19.128.65. Your company's email is username@theBest.com

   (a) What Resource Records (RRs) do you need to provide to DNS Registrar (Network Solutions).
   (b) What RRs do you need to put in your company's Authoritative DNS Servers.

Assignment Guidelines:

*To be done in **groups of 3 or 4 students; NO individual submission**.*

*To be handwritten on plain **A4 size paper**; all group members to contribute in writing the assignment.*

*Assignment to have a separate **title page,** with **Course Name**, **Assign Num**, **Section**, **Student Name**, **Reg Num**.*

*If **on-campus classes,** Assign to be submitted in **hard copy***

*If **on-line class,** then **scanned copy** to be submitted on **UCP Portal**; only one group member to submit.*

*Assignment will **NOT be accepted after due date.***